

YOUNGJIN HONG

(763)388-7740 | hong0745@umn.edu | [LinkedIn](#) | [Google Scholar](#) | [Website](#)

RESEARCH INTERESTS

Vision-Language-Action Models, Multimodal Representation Learning, Generative Policies for Robotic Manipulation.

EDUCATION

University of Minnesota (UMN)

Ph.D. Electrical Engineering

Minneapolis, MN, USA

Sep. 2024 – Present

Sungkyunkwan University (SKKU)

M.S. in Mechanical Engineering

B.S. in Mechanical Engineering

Suwon, Korea

Mar. 2022 – Feb. 2024

Mar. 2016 – Feb. 2022

WORK EXPERIENCE

Robot Research Assistant [\[Website\]](#)

Choice Robotics Lab, University of Minnesota

Minneapolis, MN

Sep. 2024 – Present

- Conduct research on multimodal reasoning and generative refinement models for large-scale robot manipulation.
- Train and evaluate multimodal robotics models on MSI GPU clusters, using SLURM for large-scale experimentation, job scheduling, and resource management.
- Developing a VLM-based multi-person attribute-conditioned tracking system for real-time visual monitoring.

Robot Research Engineer [\[Website\]](#)

Hanwha Aerospace, Manned-Unmanned Teaming (MUM-T) Research Center

Seoul, Korea

Jan. 2024 – Jul. 2024

- Built Unmanned Ground Vehicle (UGV) simulation environments with Isaac Sim for vision-based autonomy.
- Contributed to a defense proposal for next-generation unmanned ground vehicle technologies.
- Performed field testing and piloting of an unmanned ground vehicle autonomy prototype.

SELECTED PROJECTS

Generative Residual Action Refinement for Manipulation (Ongoing Research)

Oct. 2025 – Present

- Building a Flow Matching-based generative refinement model that learns residual corrections to improve general-purpose manipulation policies.
- Benchmarking refinement architectures (Mamba-driven Flow Matching, CVAE) across diverse manipulation tasks.
- Preliminary results show improved success rates over base policies.

Language-Action Cycles for Learning Manipulation Tasks

Jun. 2025 – Oct. 2025

- Proposed a bidirectional language-action cycle that jointly learns action-to-language and language-to-action mappings for scalable behavior acquisition.
- Designed a semantic consistency verifier for filtering self-generated samples, enabling reliable self-improvement.

Vision-Based Planar Pushing for Service Robotics (SKKU)

Jul. 2022 – Dec. 2024

- Developed a learning-based 2D non-prehensile motion planner for dishware with unknown physical properties.
- Automated synthetic train data using Isaac Gym simulator to train the push planning network.

SELECTED PUBLICATIONS / PRESENTATIONS

Tian Xie*, **Youngjin Hong*** (*equal contribution) et al., "LIBERO-LQ: A Latency- and Quality-Aware Benchmark for Vision-Language-Action Models", Under review at *ICML 2026* [\[Website\]](#)

Youngjin Hong*, Houjian Yu* (*equal contribution) et al., "LACY: A Vision-Language Model-based Language-Action Cycle for Self-Improving Robotic Manipulation", *ICRA 2026*. [\[Website\]](#) [\[Paper\]](#)

Mingen Li, Houjian Yu, Yixuan Huang, **Youngjin Hong**, Hantao Ye, Changhyun Choi. "Hierarchical DLO Routing with Reinforcement Learning and In-Context Vision-language Models", *ICRA 2026*. [\[Website\]](#) [\[Paper\]](#)

SKILLS

Programming: Python (proficient), MATLAB, C++ (familiar) | **Tools/Infrastructure:** Docker, Git, Linux, SLURM, GPU Clusters

Deep Learning:

PyTorch, HuggingFace, Generative Models, **Vision-Language Models (VLMs)**, **Vision-Language-Action (VLA) Models**

Robotics: ROS, Isaac Gym, MuJoCo, CoppeliaSim

Courses: Robot Vision, Deep Learning, Reinforcement Learning, Advanced Topics in Generative AI, Optimization Theory